May 13, 2026

Editors

SIAM Journal on Matrix Analysis and Applications

Dear Editors,

Please consider the manuscript entitled "A Channel-Conditioned Perturbation Theorem for Hidden Schur-Complement Resolvents" for publication in SIAM Journal on Matrix Analysis and Applications.

The manuscript develops a perturbation theory for restricted hidden inverse channels. Its central contribution is a channel-conditioned perturbation bound for the inverse operator transported by singular subspaces, together with a sharpness construction, a canonical normal form, an independent-variation theorem for scalar gain, channel conditioning, access, and visibility, and an impossibility theorem showing that composite input-output maps cannot certify hidden channel geometry.

The work is intended for readers in matrix analysis, numerical linear algebra, perturbation theory, model reduction, robust control, and nonnormal dynamics. It distinguishes the proposed channel condition number from classical scalar gain, Wedin/Davis--Kahan subspace perturbation bounds, and observed input-output transfer summaries. The included finite-dimensional benchmark and Orr--Sommerfeld/Squire stress test are generated by the accompanying Python appendix.

The manuscript is original, is not under consideration elsewhere, and all authors have approved its submission. To preserve anonymous review, author-identifying information has been anonymized in the manuscript file where appropriate.

Thank you for your consideration.

Sincerely,

The authors